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Refrigeration and Air-Conditioning

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Syllabus

Module I

Principles of refrigeration-unit of refrigeration - capacity - Coefficient of Performance - refrigeration systems: Carnot refrigeration cycle - Steam jet refrigeration - Thermoelectric refrigeration - vortex tube - pulse tube - air refrigeration cycle boot strap & boot strap evaporating cooling - thermodynamic analysis of Bell- Coleman cycle.

Module II

Vapour compression system - theoretical and practical cycles - simple and multi pressure systems - thermodynamic analysis - vapour absorption system - principle of operation of aqua - ammonia and lithium bromide - water systems - Electrolux system - comparison between vapour compression and absorption systems - refrigerants - thermodynamic, physical and chemical properties of refrigerants, environment friendly refrigerants and its properties, selection criteria of refrigerants.

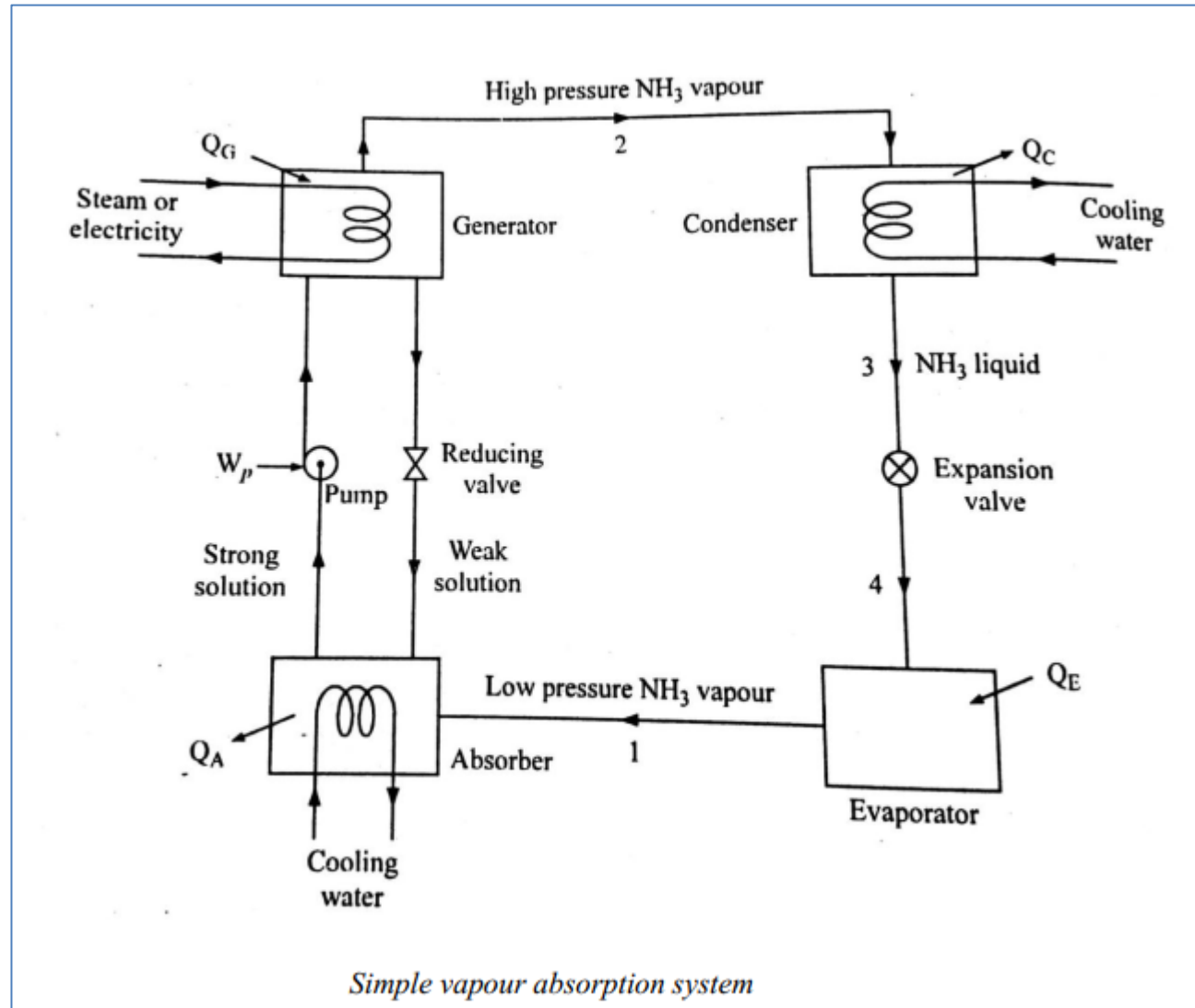
Vapor Absorption Systems

VAPOUR ABSORPTION REFRIGERATION SYSTEMS



- The vapour absorption system uses heat energy, instead of mechanical energy as in vapour compression systems, in order to change the conditions of the refrigerant required for the operation of the refrigeration cycle
- Compressor is replaced by an absorber, a pump, a generator and a pressure reducing valve
- These components in vapour absorption system perform the same function as that of a compressor in vapour compression system

VAPOUR ABSORPTION REFRIGERATION SYSTEMS



VAPOUR ABSORPTION REFRIGERATION SYSTEMS



- Vapour refrigerant from the evaporator is drawn into an absorber where it is absorbed by the weak solution of the refrigerant forming a strong solution.
- This strong solution is pumped to the generator where it is heated by some external source
- Low pressure ammonia vapour leaving the evaporator enters the absorber where it is absorbed by the cold water in the absorber
- The water has the ability to absorb very large quantities of ammonia vapour and the solution thus formed, is known as aqua-ammonia

VAPOUR ABSORPTION REFRIGERATION SYSTEMS



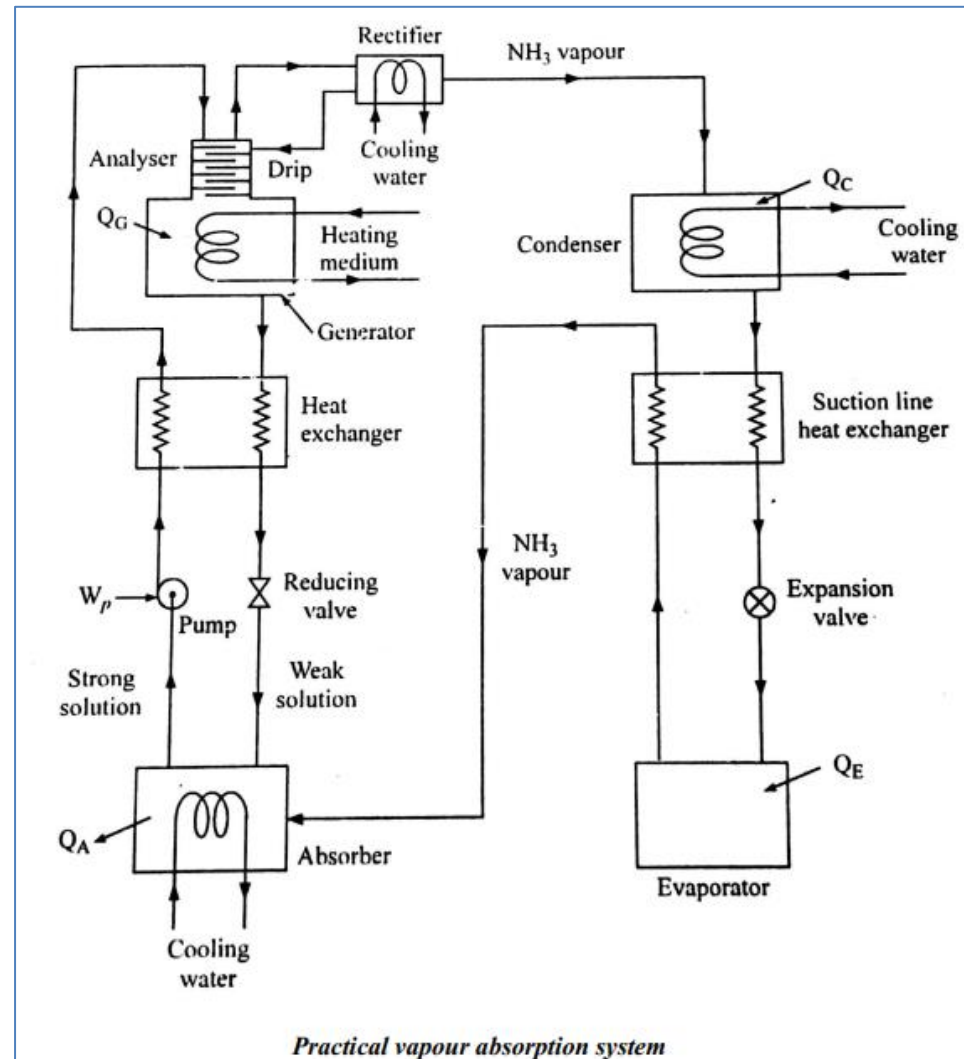
- The absorption of ammonia vapour in water lowers the pressure in the absorber which in turn draws more ammonia vapour from the evaporator and thus raises the temperature of solution
- Some form of cooling arrangement (usually water cooling) is employed in the absorber to remove the heat of solution evolved there. This is necessary in order to increase the absorption capacity of water, because at higher temperature water absorbs less ammonia vapour
- The strong solution thus formed in the absorber is pumped to the generator by the liquid pump. The pump increases the pressure of the solution up to 10 bar

VAPOUR ABSORPTION REFRIGERATION SYSTEMS



- The strong solution of ammonia in the generator is heated by some external source such as gas or steam. During the heating process, the ammonia vapour is driven off the solution at high pressure leaving behind the hot weak ammonia solution in the generator
- This weak ammonia solution flows back to the absorber at low pressure after passing through the pressure reducing valve. The high pressure ammonia vapour from the generator is condensed in the condenser to high pressure liquid ammonia.
- This liquid ammonia is passed to the expansion valve through the receiver and then to the evaporator

ACTUAL VAPOUR ABSORPTION REFRIGERATION SYSTEMS



ACTUAL VAPOUR ABSORPTION REFRIGERATION SYSTEMS



- These accessories help to improve the performance and working of the plant
1. **Analyser**:- Removes unwanted water vapours from ammonia vapours in generator. The strong solution from the absorber and the aqua from the rectifier are introduced at the top of the analyser and flow downward over the trays and into the generator. In this way, considerable liquid surface area is exposed to the vapour rising from the generator. The vapour is cooled and most of the water vapour condenses, so that mainly ammonia vapour leaves the top of the analyser

ACTUAL VAPOUR ABSORPTION REFRIGERATION SYSTEMS



2. **Rectifier:-** In case the water vapours are not completely removed in the analyser, a closed type vapour cooler called rectifier (also known as dehydrator) is used. It is generally water cooled and may be of the double pipe, shell and coil or shell and tube type. Its function is to cool further the ammonia vapours leaving the analyser so that the remaining water vapours are condensed.
3. **Heat exchanger:-** The heat exchanger provided between the pump and the generator is used to cool the weak hot solution returning from the generator to the absorber. The heat removed from the weak solution raises the temperature of the strong solution leaving the pump and going to analyser and generator.

VCRS vs VARS



- In the vapour absorption system, the only moving part of the entire system is a pump which has a small motor. Thus, the operation of this system is essentially quiet and is subjected to little wear. The vapour compression system of the same capacity has more wear, tear and noise due to moving parts of the compressor
- The vapour absorption system uses heat energy to change the condition of the refrigerant from the evaporator. The vapour compression system uses mechanical energy to change the condition of the refrigerant from the evaporator

VCRS vs VARS



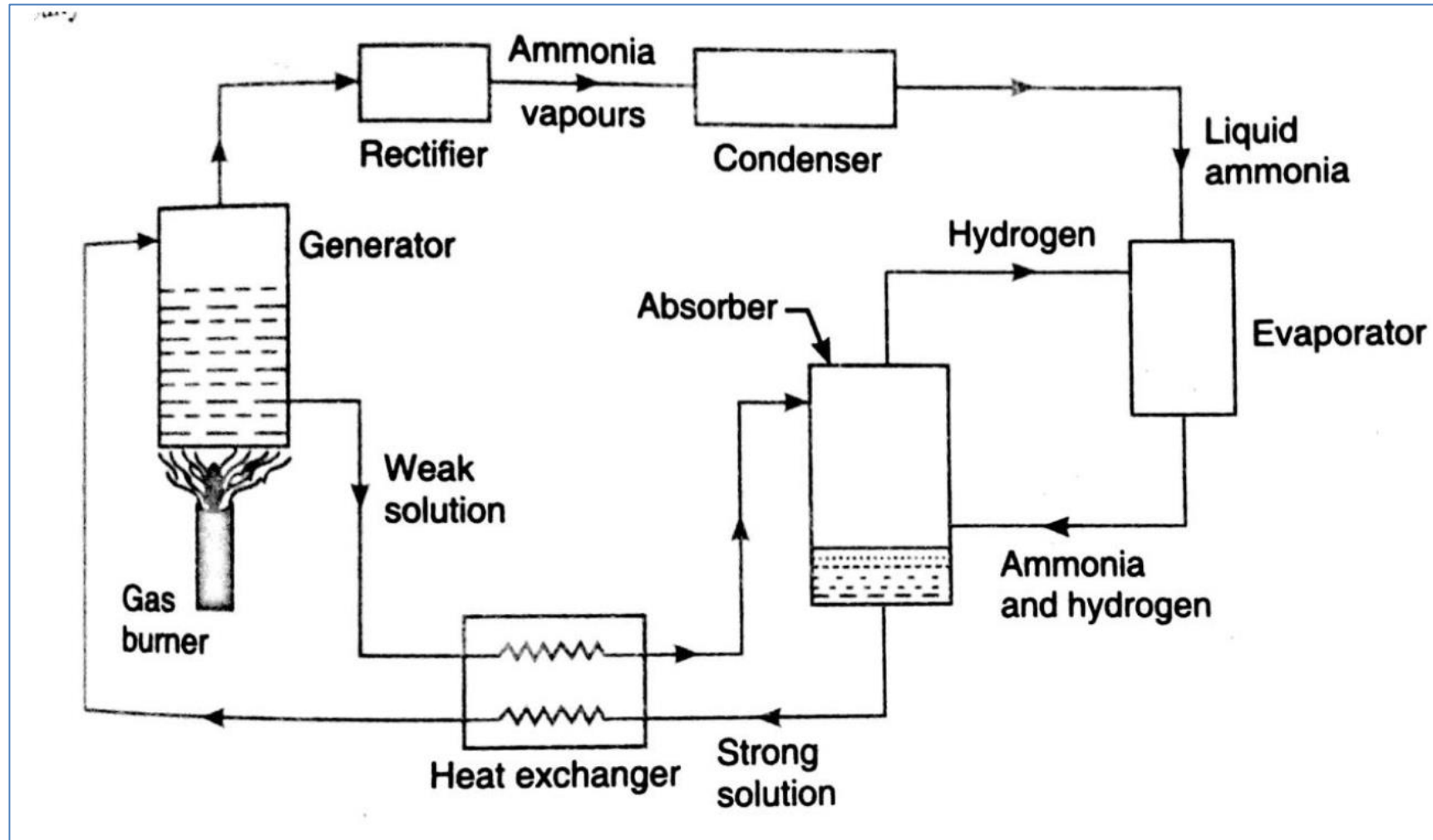
- The vapour absorption systems are usually designed to use steam, either at high pressure or low pressure. The exhaust steam from furnaces and solar energy may also be used. Thus this system can be used where the electric power is difficult to obtain or is very expensive
- The vapour absorption systems can operate at reduced evaporator pressure and temperature by increasing the steam pressure to the generator, with little decrease in capacity. But the capacity of vapour compression system drops rapidly with lowered evaporator pressure
- The vapour absorption systems can be built in capacities well above 1000 tonnes of refrigeration each which is the largest size for single compressor units

DOMESTIC ELECTROLUX (AMMONIA HYDROGEN) or THREE FLUID REFRIGERATOR



- Idea was first developed by Electrolux company
- Three fluids
 - Ammonia
 - Hydrogen
 - Water
- Ammonia:- As refrigerant
- Hydrogen:- used to increase the rate of evaporation of the liquid ammonia passing through the evaporator
- Water:- Used as a solvent because it has the ability to absorb ammonia readily

DOMESTIC ELECTROLUX (AMMONIA HYDROGEN) or THREE FLUID REFRIGERATOR



DOMESTIC ELECTROLUX (AMMONIA HYDROGEN) or THREE FLUID REFRIGERATOR



- The hydrogen gas which is being fed to the evaporator permits the liquid ammonia to evaporate at a low pressure and temperature according to Dalton's principle. During the process of evaporation, the ammonia absorbs latent heat from the refrigerated space and thus produces cooling effect
- The mixture of ammonia vapour and hydrogen is passed to the absorber where ammonia is absorbed in water while the hydrogen rises to the top and flows back to the evaporator.

DOMESTIC ELECTROLUX (AMMONIA HYDROGEN) or THREE FLUID REFRIGERATOR

Advantages:

1. No pump or compressor work is required.
2. The system is very reliable and maintenance cost is less.
3. No lubrication problem, no wear and tear.
4. Completely leak proof and Noiseless
5. No chance of pressure unbalancing and no need of valves.
6. Easy control, simply by controlling heat input.

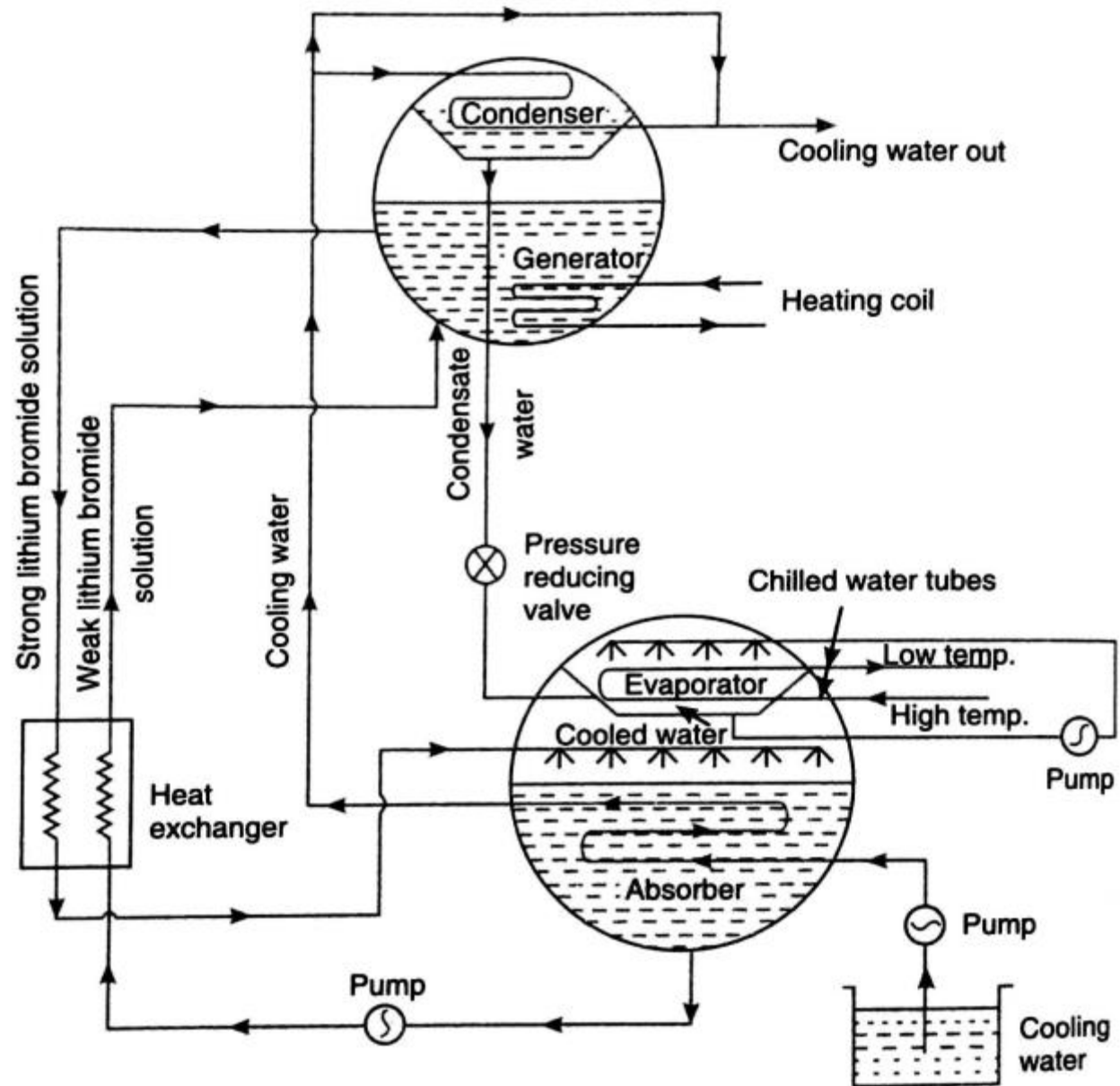
Disadvantages:

1. More complicated in construction and working.
2. C.O.P very low
3. The major disadvantage of this type of refrigerator is that if it is spoiled once, it cannot be repaired and has to be replaced fully.

Lithium Bromide Absorption Refrigeration System



- Uses a solution of Lithium bromide in water
- Water:- Refrigerant
- LiBr:- highly hygroscopic salt, as an absorbent
- The Lithium bromide solution has a strong affinity for water vapour because of its very low vapour pressure
- Absorber and the evaporator are placed in one shell which operates at the same low pressure of the system. The generator and condenser are placed in another shell which operates at the same high pressure of the system



Lithium-Bromide absorption refrigeration system

Thank You